# MF RECOGNITION IS $\Pi^0_2$-COMPLETE

SAUERS

A countable group is MF if it admits finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate non-identity elements in operator norm. We show that deciding whether a finite presentation defines an MF group is $\Pi^0_2$-complete. So it is impossible to construct an algorithm that decides from a finite presentation whether the group it defines is MF, and impossible even to enumerate the presentations of MF groups, or those of non-MF groups. A presentation defines an MF group if and only if, for every accuracy, every finite word challenge has either a finite word-problem witness or an explicitly coded rational unitary microstate with decidable norm certificates. We also construct, computably from any program, a finite presentation that defines an MF group if and only if the program halts on infinitely many inputs. If the program halts on only finitely many inputs, the presented group contains the finitely presented non-MF group of [NMF, Theorem C]; if it halts on infinitely many inputs, the group is MF by a permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona.

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## 1. Introduction

Not every countable group is MF: *Non-MF Groups* [NMF] constructs a finitely generated simple group $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ with property (T) such that every homomorphism from $H$ to an MF group is trivial. We turn to the recognition problem for finitely presented groups.

**Theorem A** (MF recognition is $\Pi^0_2$-complete)**.** *Deciding whether a finite presentation defines an MF group is $\Pi^0_2$-complete, and the complementary problem is $\Sigma^0_2$-complete. In particular, it is impossible to construct an algorithm that decides from a finite presentation whether the group it defines is MF, or even one that enumerates the presentations of MF groups or those of non-MF groups. There is a computable map $e \mapsto \widehat{R}_e$ such that $\widehat{R}_e$ presents*

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*an MF group if and only if the $e$-th partial computable function has infinite domain.*

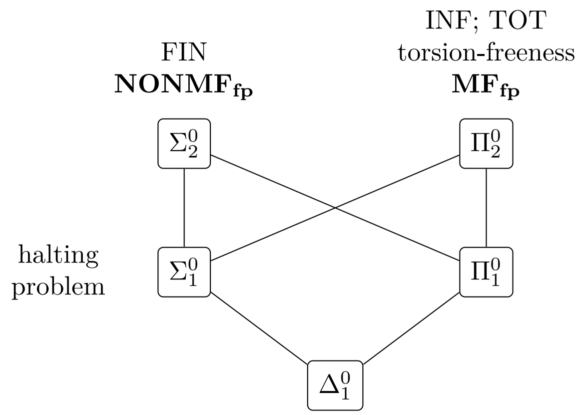


Figure 1: The arithmetical hierarchy. Lines indicate inclusions, and each listed problem is complete at its level. The two sets from Theorem A appear in bold. $\mathrm{MF_{fp}} \in \Pi_2^0$ by finite certificates (Section 3). The family $\widehat{R}_e$ reduces INF to $\mathrm{MF_{fp}}$ and FIN to $\mathrm{NONMF_{fp}}$ (Section 5).

Encode a finite challenge as a pair $(W, k)$, where $W$ is a finite list of words and $k$ is a reciprocal precision index. An answer either supplies a word of $W$ together with a finite word-problem search witness proving that it is trivial, or supplies Gaussian-rational unitary matrices for the generators together with finite certificates that every relator has norm defect less than $1/(k+1)$ and every word of $W$ has norm displacement greater than $1/3$. All checks are primitive recursive. A finite presentation defines an MF group if and only if every coded challenge has an accepted coded answer. So $\mathrm{MF_{fp}} \in \Pi_2^0$ and $\mathrm{NONMF_{fp}} \in \Sigma_2^0$ (Section 3).

Given a program index $e$, we construct a finite presentation $\widehat{R}_e$ in Section 5. If the $e$-th partial function has finite domain, then the presented group contains the finitely presented non-MF group $E$ of [NMF, Theorem C] and is not MF. If its domain is infinite, the edge isomorphism in the defining HNN extension is implemented by a unitary of a norm matrix corona, and the presented group is MF by Theorem 4.2. The two sets of Theorem A appear in bold in Figure 1.

**Relation to prior work.** Decision problems for finitely presented groups go back to Dehn, and the theorem of Adian and Rabin [Ad, Ra] shows that no Markov property (one that some finitely presented group has and that no group containing some fixed finitely presented group has) can be recognized from a finite presentation. Once a finitely presented non-MF group exists, the MF property is Markov and so, by Adian–Rabin, undecidable; Theorem A determines the exact level. A classical $\Pi^0_2$-complete recognition problem for finitely presented groups is torsion-freeness, by Lempp [Lem]. The first step of Section 5 is a recursive presentation that is trivial or isomorphic to $E$ according to whether the $e$-th domain is infinite or finite. The operator algebra enters in the passage from recursive to finite presentations: the HNN extensions of Section 5 are chosen so that Theorem 4.2 applies to them.

## 2. MF groups

A countable group $G$ is *MF* if there are positive integers $d_n$ and maps $V_n \colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE]. Here, MF group always means this operator-norm local-approximation property. The strong-convergence convention also requires the matrix models to recover the norms of the left regular representation [GKM, Sch]. Equivalently, $G$ embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) \Big/ \bigoplus_n M_{d_n}(\mathbb{C}),$$

where $\bigoplus_n M_{d_n}(\mathbb{C})$ is the $c_0$-direct sum of the sequences with $\|x_n\| \to 0$. A separable $C^*$-algebra is called MF if it embeds as a $C^*$-subalgebra of a norm matrix corona [BK]. Restricting models to a subgroup shows that subgroups of MF groups are MF.

Throughout, the binary Leavitt algebra $L_{\mathbb{F}_2}(1,2)$ is the unital $\mathbb{F}_2$-algebra generated by $s_0, s_1, t_0, t_1$ subject to $t_i s_j = \delta_{ij}$ and $s_0 t_0 + s_1 t_1 = 1$, and $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ is the subgroup of $\mathrm{GL}_{12}(L_{\mathbb{F}_2}(1,2))$ generated by the elementary matrices $e_{ij}(a) = 1 + aE_{ij}$, $i \neq j$. By [NMF, Theorem B], every homomorphism from $H$ to an MF group is trivial; in particular $H$ is not MF.

## 3. Locality and finite certificates

The MF property is defined through sequences of models on all of $G$, but it is determined by finite data with a fixed separation constant. The reformulation is due to Korchagin [Kor].

The following finite-domain formulation is derived from Korchagin [Kor, Proposition 7].

**Lemma 3.1** (locality with fixed separation)**.** *A countable group $G$ is MF if and only if for every finite set $F \subseteq G$ containing $1$ and every $\varepsilon > 0$ there are $d \geq 1$ and a map $V \colon F \to \mathcal{U}(d)$ with $V(1) = 1$ such that*

$$\|V(gh) - V(g)V(h)\| \leq \varepsilon \quad \textit{whenever } g, h, gh \in F,$$

$$\|V(g) - 1\| \geq 1 \quad \textit{for } g \in F \setminus \{1\}.$$

*Proof.* Maps for an exhaustion $(F_n, 1/n)$ of $G$, extended by 1 outside $F_n$, form an MF model. Conversely, Korchagin's amplification — replacing $V_n$ by an iterated adjoint representation $\mathrm{Ad}^k \circ V_n$ — raises the separation of a fixed nonidentity element to $\sqrt{2}$ while multiplying the defects by a constant [Kor, Proposition 5], and a block direct sum of late amplified models, one summand for each $g \in F \setminus \{1\}$, gives the stated map [Kor, Proposition 7]. ☐

We use the Kleene–Mostowski arithmetical hierarchy in its standard form [So]: a set $A \subseteq \mathbb{N}$ is $\Sigma^0_1$ if $A = \{x : \exists y\, R(x,y)\}$ for a decidable relation $R$, it is $\Pi^0_1$ if its complement is $\Sigma^0_1$, and $\Sigma^0_2$ and $\Pi^0_2$ are the classes $\{x : \exists y\, \forall z\, R(x,y,z)\}$ and $\{x : \forall y\, \exists z\, R(x,y,z)\}$ with $R$ decidable. Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write $G_P$ for the group presented by the code $P$. Let $\mathrm{MF}_{\mathrm{fp}}$ be the set of codes $P$ such that $G_P$ is MF, and $\mathrm{NONMF}_{\mathrm{fp}}$ its complement. Korchagin's Proposition 8 [Kor] is the finite-presentation form of Lemma 3.1; the certificates below add normal-closure witnesses for the trivial words, and the relation becomes decidable.

**Proposition 3.2** (finite certificates). *There is a decidable relation $C(P,n,c)$ on triples of natural numbers such that a code $P$ lies in $\mathrm{MF}_{\mathrm{fp}}$ if and only if for every $n$ there is $c$ with $C(P,n,c)$. Consequently $\mathrm{MF}_{\mathrm{fp}} \in \Pi^0_2$ and $\mathrm{NONMF}_{\mathrm{fp}} \in \Sigma^0_2$.*

*Proof.* Let $P = \langle x_1, \dots, x_s \mid r_1, \dots, r_m \rangle$. A *challenge* is a pair $(W,k)$, where $W$ is a finite list of words in the generators and $k \in \mathbb{N}$. Say that the challenge is answered if either

(1) some $w \in W$ represents 1 in $G_P$, or
(2) there are unitary matrices $U_1, \dots, U_s$ such that

$$\|r_i(U) - 1\| < \frac{1}{k+1} \quad (1 \leq i \leq m), \qquad \|w(U) - 1\| > \frac{1}{3} \quad (w \in W).$$

The local characterization of MF gives

$$G_P \text{ is MF} \iff \text{every finite challenge } (W,k) \text{ is answered.} \tag{1}$$

For the forward implication, apply Lemma 3.1 to the finite set represented by $W$ and to a sufficiently fine multiplicative scale; if one of the challenged words is trivial, use the first answer, and otherwise retain slack in the two strict inequalities. Conversely, choose a word representative of each element of $G_P$. Applying the second answer to finite lists of pairwise quotient words gives uniformly separated finite models. The relator bounds, together with a fixed finite normal-closure expression for each multiplication word, make their multiplicative defects tend to zero. This is the MF condition.

We now replace the existential matrix statement by finite discrete data. A *void answer* consists of a word $w \in W$ and a finite search witness exhibiting

$w$ as a product of conjugates of the relators and their inverses. A *matrix answer* consists of a positive dimension, Gaussian-rational matrices for the generators, one upper-norm certificate for each relator, and one lower-norm certificate for each word of $W$. The matrix checker verifies exact unitarity. For upper bounds it uses the power/Frobenius estimates for $X^*X$; for lower bounds it uses a coded rational vector. So every field is finite data and the complete answer checker is primitive recursive. Every strict matrix answer in (1) has such a coded rational answer: the Cayley transform $(1+ia)(1-ia)^{-1}$ of a Gaussian-rational Hermitian matrix $a$ is a Gaussian-rational unitary, and these unitaries are dense in $\mathcal{U}(d)$; soundness of the norm certificates gives the converse.

Fix total computable decoders $n \mapsto (W_n, k_n)$ for challenges and $c \mapsto a_c$ for answers, both surjective, and let $C(P, n, c)$ mean that the Boolean answer checker accepts $a_c$ for $(W_n, k_n)$. Then $C$ is primitive recursive and (1) becomes

$$P \in \mathrm{MF}_{\mathrm{fp}} \iff \forall n\, \exists c\; C(P, n, c).$$

Thus $\mathrm{MF}_{\mathrm{fp}} \in \Pi_2^0$, and taking complements gives $\mathrm{NONMF}_{\mathrm{fp}} \in \Sigma_2^0$. □

## 4. HNN extensions with a corona conjugator

In [NMF] we prove non-approximability using one-sided conjugation in a norm matrix corona. If the edge isomorphism of an HNN extension is implemented by a unitary of a norm matrix corona, then the HNN extension is MF. For $\theta = \mathrm{id}$ this is close to Shulman's theorem on central HNN extensions [Sh]; the version below assumes a tracial state vanishing off the identity and allows a nontrivial edge isomorphism. The corona unitary implements the edge isomorphism, while the trace and the left regular representation prove that the group embeds in the resulting algebra.

A countable group $G$ *admits a tracial MF realization* $(A, \rho, \tau)$ if $A$ is a separable unital MF $C^*$-algebra, $\rho\colon G \to \mathcal{U}(A)$ is a homomorphism, and $\tau$ is a tracial state on $A$ with $\tau(\rho(g)) = 0$ for every $g \neq 1$. Since $\tau(1) = 1$, such a group embeds in $\mathcal{U}(A)$, so it is MF. Only the vanishing of $\tau$ off the identity is used below, and the constructions below produce tracial states. To embed an arbitrary norm matrix corona $\mathcal{Q}_{\mathbf{d}}$ in $\prod_k M_k(\mathbb{C})/\bigoplus_k M_k(\mathbb{C})$, choose a strictly increasing sequence $(m_n)$ with $m_n \geq d_n$, embed $M_{d_n}(\mathbb{C})$ as a corner of $M_{m_n}(\mathbb{C})$, and put zero in the unused coordinates. The resulting map on reduced products is an isometric $*$-homomorphism, so the MF algebras of Section 2 are exactly the separable $C^*$-algebras that embed in the latter corona, as in [BK, Sh].

**Lemma 4.1** (residually finite groups)**.** *Every countable residually finite group admits a tracial MF realization.*

*Proof.* Choose finite-index normal subgroups $N_1 \geq N_2 \geq \cdots$ with trivial intersection, and let $\lambda_k$ be the left regular representation of $G/N_k$, of dimension $d_k$. In the corona $\mathcal{Q}_{\mathbf{d}}$ the map $\rho(g) = [(\lambda_k(g))_k]$ is a homomorphism, and $\tau([(x_k)_k]) = \lim_\omega \operatorname{tr}_{d_k}(x_k)$, for a free ultrafilter $\omega$, is a tracial state with $\tau(\rho(g)) = 0$ for $g \neq 1$, since $g \notin N_k$ for all large $k$. The $C^*$-algebra generated by $\rho(G)$ and the unit of the corona is separable and unital, and the triple with $\rho$ and $\tau$ is the required realization. $\square$

Let $G$ be a countable group, let $S \leq G$ be a subgroup, and let $\theta\colon S \to G$ be an injective homomorphism. The HNN extension

$$R = \langle G, t \mid tst^{-1} = \theta(s)\ (s \in S)\rangle \tag{2}$$

contains $G$ by Britton's lemma [Bri].

**Theorem 4.2** (HNN permanence with a corona conjugator). *In the situation of (2), suppose that $G$ admits a tracial MF realization $(A, \rho, \tau)$, and that for some norm matrix corona $\mathcal{Q}$ there are an injective $*$-homomorphism $\iota\colon A \to \mathcal{Q}$ and a unitary $W \in \mathcal{Q}$ with*

$$W\,\iota\rho(s)\,W^* = \iota\rho(\theta(s)) \qquad (s \in S).$$

*Then $R$ admits a tracial MF realization. In particular, $R$ is MF.*

*Proof.* Write $D = C^*(\iota\rho(G)) \subseteq \mathcal{Q}$, a separable $C^*$-algebra with unit $p = \iota(1)$, and let $B_0 = C^*(\iota\rho(S))$ and $B_1 = C^*(\iota\rho(\theta(S)))$, both containing $p$. Conjugation by $W$ restricts to a $*$-isomorphism $\Theta\colon B_0 \to B_1$ carrying $\iota\rho(s)$ to $\iota\rho(\theta(s))$. Let $U$ be the quotient of the full unital free product $D * C(\mathbb{T})$ by the closed ideal generated by the elements $ubu^* - \Theta(b)$, $b \in B_0$, where $u$ is the canonical unitary generator of $C(\mathbb{T})$. We write $d$ and $u$ for the images of $d \in D$ and $u$ in $U$. By construction, $U$ has the following universal property: whenever $\pi\colon D \to E$ is a unital $*$-homomorphism into a unital $C^*$-algebra and $v \in \mathcal{U}(E)$ satisfies $v\pi(b)v^* = \pi(\Theta(b))$ for $b \in B_0$, there is a unique $*$-homomorphism $U \to E$ extending $\pi$ and sending $u$ to $v$. We first embed $U$ in an amalgamated free product, which Shulman's criterion shows to be MF, and then construct the required tracial state on the copy of $R$ inside $U$.

*Step 1: $U$ embeds in an amalgamated free product.* Let $C = B_0 \oplus B_1$, $A_1 = M_2(D)$, and $A_2 = M_2(B_0)$, with the unital inclusions

$$\iota_A(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} \in A_1, \qquad \iota_B(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & \Theta^{-1}(c_1) \end{pmatrix} \in A_2,$$

and let $P = A_1 *_C A_2$ be the full amalgamated free product, in which $A_1$ and $A_2$ embed [Sh, Section 2.1]. Ueda proved that the universal HNN extension is a full corner of this amalgamated free product [Ue, Proposition 2.4]: for the projection $e = \iota_A(1_D, 0) = \iota_B(1_D, 0)$ there is an isomorphism of $U$ onto $ePe$ carrying $d \in D$ to $\operatorname{diag}(d, 0)$ and the unitary $u$ to the partial isometry $e_{12}f_{21}$ built from the matrix units $e_{ij}$ of $A_1$ and $f_{ij}$ of $A_2$. In particular $U$ embeds in $P$.

*Step 2: $P$ is MF.* By Shulman's amalgamated-free-product criterion [Sh, Theorem 16], the full amalgamated free product $A_1 *_C A_2$ of separable $C^*$-algebras is MF as soon as there are injective $*$-homomorphisms $\varphi_A$ and $\varphi_B$ of $A_1$ and $A_2$ into one norm matrix corona with $\varphi_A \circ \iota_A = \varphi_B \circ \iota_B$.

Identify $M_2(\mathcal{Q})$ with the norm matrix corona of the doubled dimension sequence, and let $\varphi_A \colon M_2(D) \to M_2(\mathcal{Q})$ be the entrywise inclusion and $\varphi_B \colon M_2(B_0) \to M_2(\mathcal{Q})$ the entrywise inclusion followed by conjugation by $\operatorname{diag}(1, W)$. Both are injective, and on $C$,

$$\varphi_B(\iota_B(c_0, c_1)) = \begin{pmatrix} c_0 & 0 \\ 0 & W\Theta^{-1}(c_1)W^* \end{pmatrix} = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} = \varphi_A(\iota_A(c_0, c_1)),$$

because $\Theta$ is conjugation by $W$. So $P$ is MF, as is its $C^*$-subalgebra $ePe \cong U$.

*Step 3: the tracial state.* The functional $T = \tau \circ \iota^{-1}$ is a tracial state on $\iota(A) \supseteq D$ with $T(\iota\rho(g)) = 0$ for $g \neq 1$. In the GNS representation of $D$ for $T$, the vectors $\xi_g = \iota\rho(g)\xi_T$, $g \in G$, are orthonormal, because $\langle \xi_g, \xi_h \rangle = T(\iota\rho(h^{-1}g))$. Their closed span is invariant under $D$, and on it $\iota\rho(g)$ acts as the left regular representation $\lambda_G(g)$ of $G$. Thus there is a $*$-homomorphism $\pi \colon D \to C^*_{\mathrm{r}}(G)$ with $\pi(\iota\rho(g)) = \lambda_G(g)$. Since $G \leq R$, the space $\ell^2(R)$ is a direct sum of copies of $\ell^2(G)$ indexed by the right cosets of $G$, so $\lambda_R|_G$ is a multiple of $\lambda_G$ and the assignment $\lambda_G(g) \mapsto \lambda_R(g)$ extends to a $*$-homomorphism $C^*_{\mathrm{r}}(G) \to C^*_{\mathrm{r}}(R)$. Composing, we obtain $\sigma_0 \colon D \to C^*_{\mathrm{r}}(R)$ with $\sigma_0(\iota\rho(g)) = \lambda_R(g)$. The pair $(\sigma_0, \lambda_R(t))$ satisfies the covariance relation $\lambda_R(t)\sigma_0(b)\lambda_R(t)^* = \sigma_0(\Theta(b))$ for $b \in B_0$: on the generators $\iota\rho(s)$ both sides equal $\lambda_R(\theta(s))$, because $tst^{-1} = \theta(s)$ in $R$, and two $*$-homomorphisms that agree on generators agree on $B_0$. The universal property of $U$ gives $\sigma \colon U \to C^*_{\mathrm{r}}(R)$ with $\sigma(\iota\rho(g)) = \lambda_R(g)$ and $\sigma(u) = \lambda_R(t)$.

Define $j \colon R \to \mathcal{U}(U)$ by $j(g) = \iota\rho(g)$ for $g \in G$ and $j(t) = u$. The defining relations of (2) hold in $U$, since $u\,\iota\rho(s)\,u^* = \Theta(\iota\rho(s)) = \iota\rho(\theta(s))$, so $j$ is a homomorphism, and $\sigma \circ j = \lambda_R$ is injective, so $j$ is injective. Let $A' = C^*(j(R)) \subseteq U$, a separable unital MF $C^*$-algebra, and let $\tau' = \tau_R \circ \sigma|_{A'}$, where $\tau_R$ is the canonical trace $x \mapsto \langle x\delta_1, \delta_1 \rangle$ of $C^*_{\mathrm{r}}(R)$. Then $\tau'$ is a tracial state on $A'$ with $\tau'(j(r)) = \tau_R(\lambda_R(r)) = 0$ for $r \neq 1$. So $(A', j, \tau')$ is a tracial MF realization of $R$. $\square$

Injectivity of $j$ follows from $\sigma \circ j = \lambda_R$, not from the covariant pair $(\iota\rho, W)$, which need not be faithful; in the applications below it is not.

**Corollary 4.3** (central HNN extensions)**.** *If $G$ admits a tracial MF realization and $S \leq G$ is any subgroup, then $\langle G, t \mid [t, s] = 1\ (s \in S) \rangle$ admits a tracial MF realization.*

*Proof.* Apply Theorem 4.2 with $\theta$ the inclusion of $S$, with $\iota$ any injective $*$-homomorphism of $A$ into a norm matrix corona, and with $W = 1$. $\square$

The positive branch of the reduction below produces tracial MF realizations inside norm reduced products. For $C^*$-algebras $B_n$, $n \geq 1$, write $\prod_n B_n / \bigoplus_n B_n$ for the quotient of the bounded sequences by the sequences with $\|x_n\| \to 0$; its norm is $\limsup_n \|x_n\|$.

**Lemma 4.4** (reduced products of MF algebras)**.** *If every $B_n$ is MF, then every separable $C^*$-subalgebra of $\prod_n B_n / \bigoplus_n B_n$ is MF. Moreover, $M_k(B)$ is MF whenever $B$ is.*

The first assertion follows by diagonalization from the matrix-corona characterization of MF algebras (Blackadar–Kirchberg [BK]; see also [BO, Chapter 11]), and Korchagin uses it in this form [Kor, Proposition 6]. The second holds because $M_k$ of a norm matrix corona is the norm matrix corona of the dimension sequence multiplied by $k$.

**Lemma 4.5** (tensor synchronization)**.** *Let $\Gamma$ be a countable group with a tracial MF realization $(A_1, \rho_1, \tau_1)$, let $Q$ be a countable group with homomorphisms $\beta_n\colon Q \to B_n$ to finite groups such that every $q \neq 1$ satisfies $\beta_n(q) \neq 1$ for all large $n$, let $S \leq \Gamma$, and let $\tau\colon S \to Q$ be a homomorphism. Suppose that for every $n$ there is a homomorphism $\lambda_n\colon \Gamma \to G_n$ to a finite group with*

$$\ker(\lambda_n|_S) \leq \ker(\beta_n \circ \tau).$$

*Then there are a separable unital MF $C^*$-algebra $A$, a homomorphism $V\colon \Gamma \times Q \to \mathcal{U}(A)$, a tracial state $T$ on $A$ with $T(V(g,q)) = 0$ for $(g,q) \neq (1,1)$, and a unitary $W \in A$ with $WV(s,1)W^* = V(s,\tau(s))$ for all $s \in S$. Consequently*

$$\big\langle \Gamma \times Q,\ u \ \ \big|\ \ u(s,1)u^{-1} = (s,\tau(s))\ (s \in S)\big\rangle$$

*admits a tracial MF realization, and in particular is MF.*

*Proof.* Let $E_n$ be the image of $(\lambda_n, \beta_n)\colon \Gamma \times Q \to G_n \times B_n$, a finite group of order $k_n$, and let $L_n$ be its left regular representation on $\ell^2(E_n)$. Put $B_n' = A_1 \otimes M_{k_n}(\mathbb{C})$, which is MF by Lemma 4.4, let $E = \prod_n B_n' / \bigoplus_n B_n'$, and define

$$V(g,q) = \big[\big(\rho_1(g) \otimes L_n(\lambda_n(g), \beta_n(q))\big)_n\big], \qquad T\big([(x_n)_n]\big) = \lim_{\omega} (\tau_1 \otimes \mathrm{tr}_{k_n})(x_n),$$

for a free ultrafilter $\omega$. Each coordinate of $V$ is a homomorphism into a unitary group, so $V$ is a homomorphism, and $T$ is a tracial state on $E$ because $|(\tau_1 \otimes \mathrm{tr}_{k_n})(x_n)| \leq \|x_n\|$. If $g \neq 1$ then $\tau_1(\rho_1(g)) = 0$, so $T(V(g,q)) = 0$; if $g = 1$ and $q \neq 1$ then $\beta_n(q) \neq 1$ for all large $n$, the regular representation of a finite group has trace zero off the identity, and again $T(V(1,q)) = 0$.

The two homomorphisms $s \mapsto (\lambda_n(s), 1)$ and $s \mapsto (\lambda_n(s), \beta_n\tau(s))$ from $S$ to $E_n$ have the same kernel, namely $\ker(\lambda_n|_S)$, by the hypothesis. So their images are isomorphic subgroups of $E_n$ of the same order, isomorphic via $(\lambda_n(s), 1) \mapsto (\lambda_n(s), \beta_n\tau(s))$. The restriction of $L_n$ to a subgroup of $E_n$ is unitarily equivalent to the direct sum of as many copies of the left regular representation of that subgroup as it has right cosets, so there is a unitary $W_n \in M_{k_n}(\mathbb{C})$ with $W_n L_n(\lambda_n(s), 1) W_n^* = L_n(\lambda_n(s), \beta_n\tau(s))$ for all $s \in S$. Put $W = [(1 \otimes W_n)_n] \in E$; then $WV(s,1)W^* = V(s,\tau(s))$ for $s \in S$. Let $A$ be the $C^*$-algebra generated by $V(\Gamma \times Q)$ and $W$; it is separable and unital, and MF by Lemma 4.4. The triple $(A, V, T|_A)$ is a tracial MF realization of $\Gamma \times Q$. For the last assertion, apply Theorem 4.2 to this realization. Let $\iota$ be an injective $*$-homomorphism of $A$ into a norm matrix corona and put $p = \iota(1)$. The element $\widetilde{W} = \iota(W) + (1-p)$ is a unitary of the full corona and implements the required covariance on $\iota(A)$, so it is the required conjugator. □

## 5. The complexity of recognizing MF groups

Write $W_e$ for the domain of the $e$-th partial computable function and

$$\mathrm{INF} = \{e : W_e \text{ is infinite}\}, \qquad \mathrm{FIN} = \{e : W_e \text{ is finite}\}.$$

The set INF is $\Pi_2^0$-complete and FIN is $\Sigma_2^0$-complete [So, Chapter IV]. The construction combines a fixed finitely presented non-MF group $E$ with Theorem 4.2: it produces a computable map $e \mapsto \widehat{R}_e$ into finite presentations such that the presented group contains $E$ when $e \in \mathrm{FIN}$ and satisfies the hypotheses of Theorem 4.2 when $e \in \mathrm{INF}$, so that it is MF if and only if $e \in \mathrm{INF}$. For $e \in \mathrm{FIN}$, the group $\widehat{R}_e$ is not MF; for $e \in \mathrm{INF}$, Lemma 5.8 proves that it is MF. Table 1 lists the objects of the construction.

| | |
|---|---|
| $E$ | the finitely presented non-MF group of [NMF, Theorem C] |
| $C_e$ | recursively presented; trivial for $e \in \mathrm{INF}$, isomorphic to $E$ for $e \in \mathrm{FIN}$ |
| $Q_e = F/N_e$ | three-generated recursively presented group containing $C_e$; here $F = F(x, y, t)$ |
| $Q_+ = F/N_+$ | the value of $Q_e$ for trivial $C_e$; embeds in $F(x_1, y) \times F(x_2, t)$ |
| $\Lambda_e$; $M_e$ | finitely presented Higman host of $Q_e$; the Mihailova subgroup detecting equality in $\Lambda_e$ |
| $K_e$; $L_e$ | a product of six free groups; its finitely generated subgroup with $i(F) \cap L_e = i(N_e)$ |
| $\Gamma_e$; $S_e$ | the central HNN extension $\langle K_e, v \mid [v, \ell] = 1 \ (\ell \in L_e)\rangle$; $S_e = \langle i(F), v\, i(F)\, v^{-1}\rangle \cong F *_{N_e} F$ |
| $\alpha_e \colon S_e \to Q_e$ | $q_e$ on $i(F)$, trivial on $v\, i(F)\, v^{-1}$ |
| $\widehat{R}_e$ | the finite presentation of $\langle \Gamma_e \times Q_e,\ u \mid u(s,1)u^{-1} = (s, \alpha_e(s)) \ (s \in S_e)\rangle$ |

Table 1: The objects of the construction, in order of appearance.

**A finitely presented non-MF seed.**

**Lemma 5.1** (the finite seed)**.** *There is a fixed finite presentation $P_-$ whose group $E = G_{P_-}$ is not MF.*

*Proof.* Theorem C of [NMF] gives a finitely presented torsion-free group that equals its own MF radical, and a nontrivial group equal to its own MF radical is not MF. Fix one finite presentation code $P_-$ for this group. Any finitely presented non-MF group serves here: Korchagin proved that the MF property is closed under direct limits [Kor, Proposition 13], so every finitely generated non-MF group, for instance $H$, is a quotient of a finitely presented one. $\square$

**The group $C_e$.**

**Lemma 5.2** (the group $C_e$)**.** *There is an algorithm that computes from $e$ a recursive presentation, on a computable set of generators, of a group $C_e$ such that $C_e$ is trivial if $e \in \mathrm{INF}$ and $C_e \cong E$ if $e \in \mathrm{FIN}$.*

*Proof.* Write $P_- = \langle X \mid R\rangle$ with $X$ finite. The generators of $C_e$ are the symbols $x^{(k)}$ for $x \in X$ and $k \in \mathbb{N}$, one copy of $X$ for each *layer* $k$. Run the $e$-th program on all inputs by dovetailing, and let $m_s$ be the number of elements of $W_e$ found by stage $s$, with $m_{-1} = 0$. The relators are enumerated in stages. At stage $s$, enumerate the relators $R$ written in the layer $s$; if $m_s > m_{s-1}$, enumerate $x^{(k)} = 1$ for all $x \in X$ and all $k \leq s$; and otherwise enumerate $x^{(s)} = x^{(s+1)}$ for all $x \in X$. The enumerator is computable in $P_-$ and $e$.

If $e \in \mathrm{INF}$, then for every $k$ some stage $s \geq k$ has $m_s > m_{s-1}$, so every generator is killed and $C_e$ is trivial. If $e \in \mathrm{FIN}$, let $s_*$ be the last stage with $m_s > m_{s-1}$, or $s_* = -1$ if there is none. The layers $k \leq s_*$ are killed; the stage $s_*$ identifies nothing, so layer $s_*$ is not identified with layer $s_* + 1$; and every stage $s > s_*$ identifies layer $s$ with layer $s + 1$. So the layers $k > s_*$ are identified with one another, and $x^{(k)} \mapsto x$ for $k > s_*$, $x^{(k)} \mapsto 1$ for $k \leq s_*$, is an isomorphism $C_e \to E$ with inverse $x \mapsto x^{(s_*+1)}$. $\square$

**A three-generator embedding.** Let $F = F(x, y, t)$ be the free group on $x, y, t$, and let $y_i = x^i y x^{-i}$ for $i \in \mathbb{Z}$. For a countable group $C$ with a generating sequence $(c_i)_{i\geq 1}$, put $c_i = 1$ for $i \leq 0$ and

$$B(C) = \big\langle C * F(x, y),\ t \ \big|\ t y_i t^{-1} = c_i y_i \ (i \in \mathbb{Z})\big\rangle. \tag{3}$$

Let $Q_+ = B(1) = \langle x, y, t \mid [t, y_i] = 1 \ (i \in \mathbb{Z})\rangle$, let $q_+ \colon F \to Q_+$ be the quotient map, and let $N_+ = \ker q_+$.

**Lemma 5.3** (the three-generator embedding)**.**

*(1) $B(C)$ is an HNN extension of $C * F(x, y)$, so $C$ embeds in $B(C)$; it is generated by $x, y, t$; and a recursive presentation of $B(C)$ on $x, y, t$ is computable from a recursive presentation of $C$. We write $B(C) = F/N_C$ and $q_C \colon F \to B(C)$ for the quotient map.*

*(2) Killing $C$ induces a surjection $B(C) \to Q_+$; so $N_C \leq N_+$, with equality when $C$ is trivial.*

*(3) The assignment $x \mapsto (x_1, x_2)$, $y \mapsto (y, 1)$, $t \mapsto (1, t)$ defines an injective homomorphism $j \colon Q_+ \to P = F(x_1, y) \times F(x_2, t)$. In particular, $Q_+$ is residually finite.*

*Proof.* (1) The elements $y_i$, $i \in \mathbb{Z}$, freely generate the normal closure of $y$ in $F(x, y)$ [LS, Chapter I]. The retraction $C * F(x, y) \to F(x, y)$ killing $C$ sends $c_i y_i$ to $y_i$, so a nontrivial reduced word in the $c_i y_i$ maps to a nontrivial word in the $y_i$; so the $c_i y_i$ freely generate a free subgroup, and $y_i \mapsto c_i y_i$ is an isomorphism between two free subgroups of the base. Thus (3) is an HNN extension, and the base, and with it $C$, embeds by Britton's lemma. The relation $c_i = ty_it^{-1}y_i^{-1}$ shows that $x, y, t$ generate $B(C)$, and substituting this expression for $c_i$ into the relators of $C$ and into (3) produces a recursive presentation on $x, y, t$, uniformly in the presentation of $C$.

(2) The relators of (3) map to the relators $[t, y_i]$ of $Q_+$ when every $c_i$ is sent to 1, and the map is the identity on $x, y, t$.

(3) The relators $[t, y_i]$ are sent to $[(1,t),(x_1^i y x_1^{-i}, 1)] = 1$, so $j$ is a homomorphism. Let $N_0$ be the normal closure of $\{y, t\}$ in $Q_+$. Then $Q_+/N_0 = \langle x\rangle$ is infinite cyclic and the extension splits, so $Q_+ = N_0 \rtimes \langle x\rangle$. Rewriting the presentation of $Q_+$ over the transversal $\{x^n\}$ by the Reidemeister–Schreier method [LS, Chapter II] presents $N_0$ on the generators $y_n = x^n y x^{-n}$ and $t_m = x^m t x^{-m}$, $n, m \in \mathbb{Z}$, with relators the conjugates $x^m[t, y_n]x^{-m} = [t_m, y_{n+m}]$, that is, all commutators $[t_m, y_n]$. So $N_0 = F(y_n : n \in \mathbb{Z}) \times F(t_m : m \in \mathbb{Z})$. Now $j(y_n) = (x_1^n y x_1^{-n}, 1)$ and $j(t_m) = (1, x_2^m t x_2^{-m})$, and the elements $x_1^n y x_1^{-n}$ freely generate the normal closure of $y$ in $F(x_1, y)$, and likewise for $t$; so $j$ restricted to $N_0$ is an isomorphism onto the product of these two normal closures. Finally $j(x) = (x_1, x_2)$ has infinite order and $j(x)^k \notin j(N_0)$ for $k \neq 0$, because, for $h \in N_0$, the exponent sum of $x_1$ in $j(x^k h)$ is $k$. So $j$ is injective on $N_0 \rtimes \langle x\rangle$. Free groups are residually finite and residual finiteness passes to direct products and subgroups, so $Q_+$ is residually finite. □

For $e \in \mathbb{N}$ let $Q_e = B(C_e) = F/N_e$ with $C_e$ the group of Lemma 5.2, and write $q_e\colon F \to Q_e$. By Lemmas 5.2 and 5.3, a recursive presentation of $Q_e$ on $x, y, t$ is computable from $e$; $N_e \le N_+$; if $e \in \mathrm{INF}$ then $Q_e = Q_+$, $N_e = N_+$ and $q_e = q_+$; and if $e \in \mathrm{FIN}$ then $E \le C_e \le Q_e$.

**Detecting $N_e$ inside a finitely presented group.**

**Lemma 5.4** (Higman embedding and the Mihailova subgroup)**.** *There is an algorithm that computes from $e$ a finite presentation $\langle X_e \mid R_e\rangle$ of a group $\Lambda_e$, words $w_x, w_y, w_t \in F(X_e)$, and a finite generating set of a subgroup $M_e \le F(X_e) \times F(X_e)$, such that $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$ induces an embedding $Q_e \to \Lambda_e$, and $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$. Consequently, with*

$$K_e^0 = F \times F(X_e) \times F(X_e), \qquad i_e(f) = (f, w_f, 1), \qquad L_e^0 = F \times M_e,$$

*where $w_f$ is the image of $f$ under $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$, the map $i_e\colon F \to K_e^0$ is injective and $i_e(f) \in L_e^0$ if and only if $f \in N_e$.*

*Proof.* Higman's embedding theorem [Hig] embeds every finitely generated recursively presented group in a finitely presented group, and its proof is effective; an explicit algorithm producing the finite presentation and the embedding words from a recursive presentation is given by Mikaelian [Mik]. Apply it to the presentation of $Q_e$. Let $M_e$ be the subgroup of $F(X_e) \times F(X_e)$ generated by the pairs $(z, z)$, $z \in X_e$, and $(r, 1)$, $r \in R_e$; then $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$ by Mihailova's lemma [Mih]. The first coordinate makes $i_e$ injective, and $(f, w_f, 1) \in F \times M_e$ if and only if $w_f = 1$ in $\Lambda_e$, if and only if $q_e(f) = 1$ because the embedding is injective, if and only if $f \in N_e$. □

**The two HNN extensions.** Let $P = F(x_1, y) \times F(x_2, t)$ and $j\colon Q_+ \to P$ be as in Lemma 5.3, and put

$$K^{\mathrm{g}} = F \times P, \qquad L^{\mathrm{g}} = \{(f,\, jq_+(f)) : f \in F\},$$

and define

$$K_e = K_e^0 \times K^{\mathrm{g}}, \qquad L_e = L_e^0 \times L^{\mathrm{g}}, \qquad i\colon F \to K_e, \quad i(f) = \big(i_e(f), (f,1)\big).$$

Let

$$\text{(4)} \quad \Gamma_e = \langle K_e,\ v \ \mid\ [v,\ell] = 1\ (\ell \in L_e)\rangle, \qquad S_e = \langle i(F),\ v\, i(F)\, v^{-1}\rangle \le \Gamma_e.$$

**Lemma 5.5** (the first HNN extension)**.**

*(1) $K_e$ is a direct product of free groups of finite rank, so finitely presented and residually finite; $L_e$ is finitely generated; $i$ is injective; and $i(F) \cap L_e = i(N_e)$.*

*(2) $\Gamma_e$ is finitely presented, and a finite presentation is computable from $e$. The natural map from the amalgamated free product $F *_{N_e} F$, in which $n \in N_e$ in the first copy is identified with $n$ in the second copy, onto $S_e$, sending the first copy to $i(F)$ and the second to $v\, i(F)\, v^{-1}$, is an isomorphism.*

*(3) There is a homomorphism $\alpha_e\colon S_e \to Q_e$ with $\alpha_e(i(f)) = q_e(f)$ and $\alpha_e(v\, i(f)\, v^{-1}) = 1$ for $f \in F$.*

*Proof.* (1) $K_e = F \times F(X_e) \times F(X_e) \times F \times P$ is a product of six free groups of finite rank, so it is finitely presented, and residually finite because free groups are. The subgroup $L_e$ is generated by the basis of the first factor, the generators of $M_e$ from Lemma 5.4, and the elements $(a, jq_+(a))$ for $a \in \{x, y, t\}$, so it is finitely generated. Injectivity of $i$ is clear from the first coordinate. An element $i(f)$ lies in $L_e$ if and only if $i_e(f) \in L_e^0$ and $(f,1) \in L^{\mathrm{g}}$; the first holds if and only if $f \in N_e$ by Lemma 5.4, and the second if and only if $jq_+(f) = 1$, that is, $f \in N_+$. Since $N_e \le N_+$, the intersection is $i(N_e)$.

(2) The presentation (4) is finite once $[v,\ell] = 1$ is imposed only for the finitely many generators $\ell$ of $L_e$, and it is computable from $e$ because $K_e$, $L_e$ and $i$ are. It is an HNN extension of $K_e$ with both edge groups equal to $L_e$ and the identity as edge isomorphism. The map from $F *_{N_e} F$ to $S_e$ is well defined because $i(n) \in L_e$ commutes with $v$ for $n \in N_e$. A reduced word of the amalgamated free product alternates between elements of the two copies of $F$ lying outside $N_e$; its image is a word $i(f_0)\, v\, i(f_1)\, v^{-1}\, i(f_2) \cdots$ in which every letter between consecutive occurrences of $v^{\pm 1}$ is some $i(f_k)$ with $f_k \notin N_e$, so $i(f_k) \notin L_e$ by (1). Such a word contains no pinch $v^{\pm 1} \ell v^{\mp 1}$ with $\ell \in L_e$, so it is nontrivial in $\Gamma_e$ by Britton's lemma.

(3) Define $\alpha_e$ on $F *_{N_e} F$ by $q_e$ on the first copy and by the trivial map on the second; both kill $N_e$, so the definition is consistent on the amalgamated subgroup, and (2) identifies $F *_{N_e} F$ with $S_e$. $\square$

The second HNN extension is

$$R_e = \langle \Gamma_e \times Q_e,\ u \ \mid\ u(s,1)u^{-1} = (s, \alpha_e(s))\ (s \in S_e)\rangle, \tag{5}$$

whose edge maps $s \mapsto (s,1)$ and $s \mapsto (s, \alpha_e(s))$ are injective homomorphisms of $S_e$ into $\Gamma_e \times Q_e$; so $\Gamma_e \times Q_e$ embeds in $R_e$ by Britton's lemma. The presentation (5) lists the recursively enumerable relators of $Q_e$. Its finite replacement is

(6)

$$\widehat{R}_e = \Big\langle \Gamma_e \times F,\ u \ \Big|\ \begin{aligned} &u(i(a),1)u^{-1} = (i(a), a),\\ &u(v\, i(a)\, v^{-1}, 1)u^{-1} = (v\, i(a)\, v^{-1}, 1) \quad (a \in \{x,y,t\}) \end{aligned} \Big\rangle,$$

where $\Gamma_e \times F$ is equipped with the finite presentation of Lemma 5.5(2), the basis $x, y, t$ of $F$, and the commutation relators between the two factors.

**Lemma 5.6** (the finite presentation)**.** *The groups $\widehat{R}_e$ and $R_e$ are isomorphic by the map that is the identity on $\Gamma_e$ and on $u$ and sends $F$ onto $Q_e$ by $q_e$. A code of the finite presentation (6) is computable from $e$.*

*Proof.* Both maps $f \mapsto u(i(f),1)u^{-1}$ and $f \mapsto (i(f), f)$ are homomorphisms $F \to \widehat{R}_e$, and they agree on the basis, so $u(i(f),1)u^{-1} = (i(f), f)$ for all $f \in F$; likewise $u(v\, i(f)\, v^{-1}, 1)u^{-1} = (v\, i(f)\, v^{-1}, 1)$ for all $f$. If $n \in N_e$, then $i(n) \in L_e$ by Lemma 5.5(1), so $v\, i(n)\, v^{-1} = i(n)$ in $\Gamma_e$, and comparing the two relations gives $(i(n), n) = (i(n), 1)$, that is, $(1, n) = 1$ in $\widehat{R}_e$. So the factor $F$ of $\widehat{R}_e$ factors through $Q_e = F/N_e$, and the relations of (6) become $u(s,1)u^{-1} = (s, \alpha_e(s))$ for $s$ in the generating set $\{i(a),\ v\, i(a)\, v^{-1} : a \in \{x,y,t\}\}$ of $S_e$; both sides define homomorphisms on $S_e$, so the relations hold for all $s \in S_e$. This defines a homomorphism $R_e \to \widehat{R}_e$; conversely, the relations of (6) hold in $R_e$ after $F \to Q_e$, because $\alpha_e(i(a)) = q_e(a)$ and $\alpha_e(v\, i(a)\, v^{-1}) = 1$. The two homomorphisms are mutually inverse on generators. Computability of the code follows from Lemmas 5.4 and 5.5. ☐

**Lemma 5.7** (the case $e \in \mathrm{FIN}$)**.** *If $e \in \mathrm{FIN}$, then $E$ embeds in $\widehat{R}_e$, and $\widehat{R}_e$ is not MF.*

*Proof.* For $e \in \mathrm{FIN}$ we have $E \le C_e \le Q_e \le \Gamma_e \times Q_e \le R_e \cong \widehat{R}_e$ by Lemmas 5.2, 5.3(1), and 5.6, and Britton's lemma for (5). Restricting MF models to a subgroup shows that subgroups of MF groups are MF, and $E$ is not MF by Lemma 5.1. ☐

**Lemma 5.8** (the case $e \in \mathrm{INF}$)**.** *If $e \in \mathrm{INF}$, then $\widehat{R}_e$ admits a tracial MF realization; in particular, it is MF.*

*Proof.* Let $e \in \mathrm{INF}$, so that $Q_e = Q_+$, $N_e = N_+$, $q_e = q_+$, and $\alpha_e(i(f)) = q_+(f)$, $\alpha_e(v\, i(f)\, v^{-1}) = 1$. By Lemma 5.6 it suffices to show that $R_e$ admits a tracial MF realization, and by Lemma 4.5 applied to $\Gamma = \Gamma_e$, $Q = Q_+$, $S = S_e$ and $\tau = \alpha_e$, it suffices to produce the data of that lemma.

The group $K_e$ is residually finite by Lemma 5.5(1), so it admits a tracial MF realization by Lemma 4.1; and $\Gamma_e$, the central HNN extension of $K_e$ over $L_e$, admits one by Corollary 4.3.

The group $P = F(x_1, y) \times F(x_2, t)$ is residually finite; choose finite-index normal subgroups $P_1 \geq P_2 \geq \cdots$ of $P$ with trivial intersection, let $r_n \colon P \to C_n = P/P_n$, and put $\beta_n = r_n \circ j \colon Q_+ \to C_n$ with $j$ the embedding of Lemma 5.3(3). Since $j$ is injective and the $P_n$ decrease to the trivial group, every $h \neq 1$ has $\beta_n(h) \neq 1$ for all large $n$.

Let $G_n = (C_n \times C_n) \rtimes \langle \sigma \rangle$, where $\sigma$ of order two exchanges the coordinates. Define $\lambda_n$ on $K_e = K_e^0 \times (F \times P)$ by killing $K_e^0$ and sending $(f, p)$ to $(r_n j q_+(f),\, r_n(p))$, and put $\lambda_n(v) = \sigma$. This respects the relations of (4): the factor $L_e^0$ of $L_e$ lies in $K_e^0$ and is killed, and an element $(f, jq_+(f))$ of $L^{\mathrm{g}}$ is sent to $(r_n j q_+(f), r_n j q_+(f))$, which lies on the diagonal and commutes with $\sigma$. So $\lambda_n \colon \Gamma_e \to G_n$ is a homomorphism, and on the generators of $S_e$,

$$\lambda_n(i(f)) = (r_n j q_+(f), 1), \qquad \lambda_n(v\, i(f)\, v^{-1}) = (1, r_n j q_+(f)).$$

Let $\pi_0, \pi_1 \colon S_e \to Q_+$ be the homomorphisms that are $q_+$ on one copy of $F$ in $S_e \cong F *_{N_+} F$ and trivial on the other, as in Lemma 5.5(3); thus $\pi_0 = \alpha_e$. The displayed formulas show that $\lambda_n|_{S_e} = (r_n j \pi_0,\, r_n j \pi_1)$, both sides being homomorphisms that agree on generators. So

$$\ker(\lambda_n|_{S_e}) = \ker(r_n j \pi_0) \cap \ker(r_n j \pi_1) \leq \ker(\beta_n \circ \alpha_e),$$

which is the hypothesis of Lemma 4.5. □

*Proof of Theorem A.* By Lemma 5.6 the map $e \mapsto \widehat{R}_e$ is computable, by Lemma 5.8 the presented group is MF for $e \in \mathrm{INF}$, and by Lemma 5.7 it is not MF for $e \in \mathrm{FIN}$. So INF reduces to $\mathrm{MF}_{\mathrm{fp}}$ and FIN reduces to $\mathrm{NONMF}_{\mathrm{fp}}$ under computable many-one reductions. Since INF is $\Pi_2^0$-complete and FIN is $\Sigma_2^0$-complete, and since $\mathrm{MF}_{\mathrm{fp}} \in \Pi_2^0$ and $\mathrm{NONMF}_{\mathrm{fp}} \in \Sigma_2^0$ by Proposition 3.2, the two sets are complete at their levels. A $\Pi_2^0$-complete set is not $\Sigma_2^0$ and a $\Sigma_2^0$-complete set is not $\Pi_2^0$, so neither set is recursively enumerable, and in particular neither is decidable. □

*Remark* 5.9. The construction uses only these properties of $E$: it is finitely presented and not MF, while subgroups of MF groups are MF. So it applies to any group property that is closed under subgroups, given a finitely presented group without the property and an analogue of Lemma 4.5 for the twisted HNN extensions; the open problems below state which input is missing for soficity and for hyperlinearity.

The homomorphism $V$ of Lemma 4.5 does not extend faithfully to $R_e$ in the corona: the finite maps $\lambda_n$ are trivial on the Mihailova factor, so $W$ commutes with $V(\gamma, 1)$ for every $\gamma \in K_e^0$, although $u$ does not commute with $(\gamma, 1)$ in $R_e$. Injectivity is obtained in Theorem 4.2 from the universal algebra.

**Open problems.** We do not know whether the models of $\widehat{R}_e$ can be chosen to converge to $C^*_{\mathrm{r}}(\widehat{R}_e)$ in operator norm, so that the reduced $C^*$-algebra itself is MF; the argument produces only a tracial state on an abstract MF algebra. We do not know whether $\widehat{R}_e$ is hyperlinear for $e \in \mathrm{INF}$; placing hyperlinearity recognition at the same level would also require a finitely presented non-hyperlinear group, and none is known. We do not know whether an analogous family of finitely presented groups exists for soficity; such a construction would require a permutation-model replacement for Lemma 4.5. Finally, we do not know the complexity of MF recognition under the strong convergence convention of [GKM, Sch]: the norms $\|\lambda(w)\|$ in the left regular representation are not in general computable from a finite presentation, so Proposition 3.2 does not apply.

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